

Jihun Baek

Cyber Security Analyst

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Core Competencies

- Splunk
- Burp Suite
- Wireshark
- OWASP Top 10
- Python scripting
- AWS WAF
- AWS GuardDuty
- Docker
- FastAPI
- Supabase
- Railway

Education

Macquarie University

Major in Cyber Security

2024 - 2025

Handong Global University

Major in Computer Science

2021 - 2022

Certifications

AWS Certified Solutions Architect – **Professional**

Feb 2025

AWS Certified Solutions Architect – **Associate**

Jan 2025

Profile

AWS Certified Solutions Architect – Professional with hands-on experience in cloud infrastructure hardening, penetration testing, and SIEM log analysis. Track record across systems administration, offensive security CTF development, and ML-based threat detection.

Work Experience

Junior IT Systems Administrator at Gomaps Trading

Nov 2025 – Present

- Administered network and cloud infrastructure for a trading company, enforcing access control via Active Directory Group Policies and maintaining high service availability.
- Architected scalable cloud email delivery infrastructure, reducing monthly costs by 15% and hardening sending domains with SPF/DKIM/DMARC.

Challenge Developer & Technical Mentor at KnockOn

Feb – Oct 2025

- Engineered 10+ CTF challenges modelled on real-world CVEs (OWASP Top 10, privilege escalation, SSRF) for 3 cohorts of security trainees — averaging a 73% solve rate, indicating well-calibrated difficulty and real-world applicability.
- Delivered weekly technical mentoring on web exploitation, network traffic analysis, and vulnerability remediation.

Projects

CyberShield — Web Security SaaS Platform

Jan – Jun 2026

- Built and deployed a production SaaS (digitalcybershield.com) — FastAPI backend on Railway, Supabase PostgreSQL/Auth, Vercel frontend with 18 API endpoints, and 60 security checks across SSL, headers, DNS, open ports, and code analysis.
- Engineered parallel async domain scanner (asyncio), real-time dashboard (Supabase WebSocket), multi-channel alerts (Slack/Discord/Email), and a static code analyser detecting 14 vulnerability patterns including hardcoded secrets, SQLi, and JWT misconfigurations.

Penetration Test — LAMP Bulletin Board System Web App

Jan – Feb 2025

- Conducted black-box pentest identifying 11 vulnerabilities including critical RCE via file upload, SQLi auth bypass, stored XSS, IDOR, and session fixation.
- Produced a structured pentest report with CVSS-style severity ratings, PoC reproduction steps, and patched source code for each finding.

AI-Based Network Attack Classifier

Nov– Dec 2024

- Built a multi-class attack classifier in Python on 220,000 network payloads (Brute Force, SQLi, XSS, System CMD) using Random Forest, XGBoost, and Logistic Regression — 92.6% accuracy, Brute Force F1: 0.99.
- Engineered 9 payload features via frequency analysis of attack-tool signatures (hydra, sqlmap, arachni) on a stratified 44,000-sample test set.